

EMAIL ANOMALY DETECTION SYSTEM

MACHINE LEARNING CASE STUDY

CORPORATE EMAIL SECURITY
ANALYTICS

EXECUTIVE SUMMARY

BUSINESS PROBLEM

Corporate email systems generate thousands of events every day. Security teams struggle to detect suspicious behavior before incidents occur.

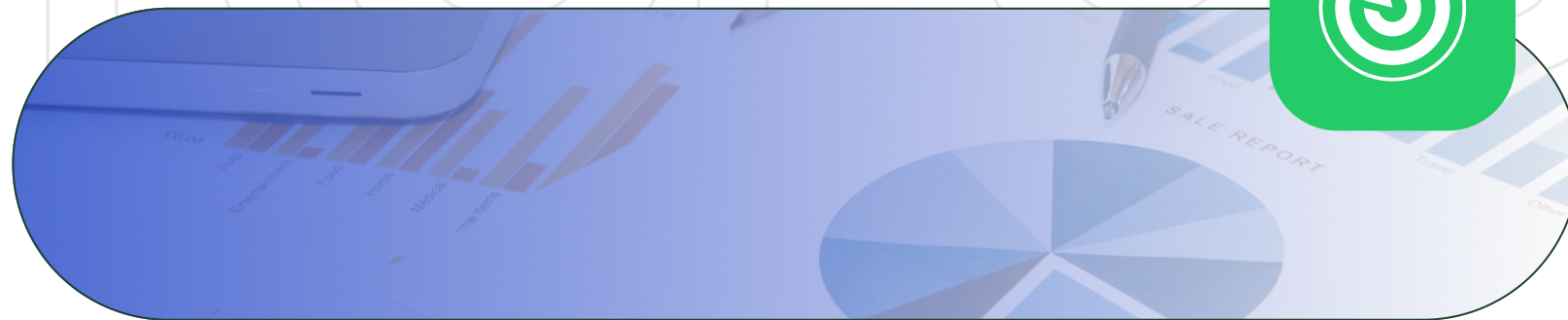
SOLUTION

Developed an end-to-end anomaly detection system using Python, PostgreSQL, Isolation Forest and Streamlit.

BUSINESS IMPACT

Allows security analysts to identify suspicious email activity, prioritize incidents and reduce manual monitoring effort.

BUSINESS CONTEXT



CORPORATE EMAIL ENVIRONMENTS ARE FREQUENT TARGETS OF PHISHING, CREDENTIAL THEFT AND INSIDER THREATS.

Challenges

- Detect unusual email activity
- Reduce false positives
- Prioritize risky events
- Support security monitoring

1050
Email Events

19
Users

Isolation
Forest
ML Model

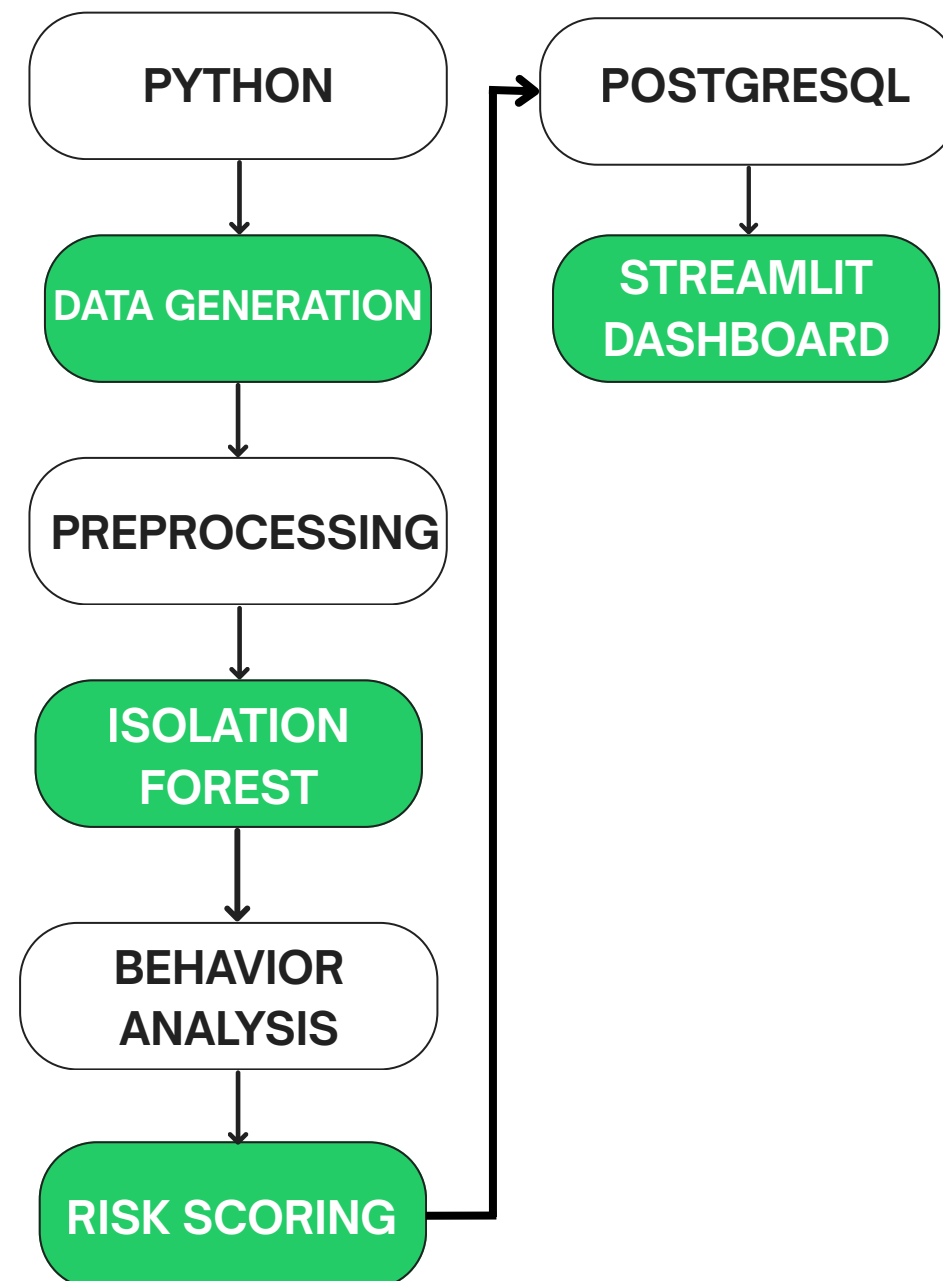
Risk
Scoring
Calculation

SYSTEM ARCHITECTURE

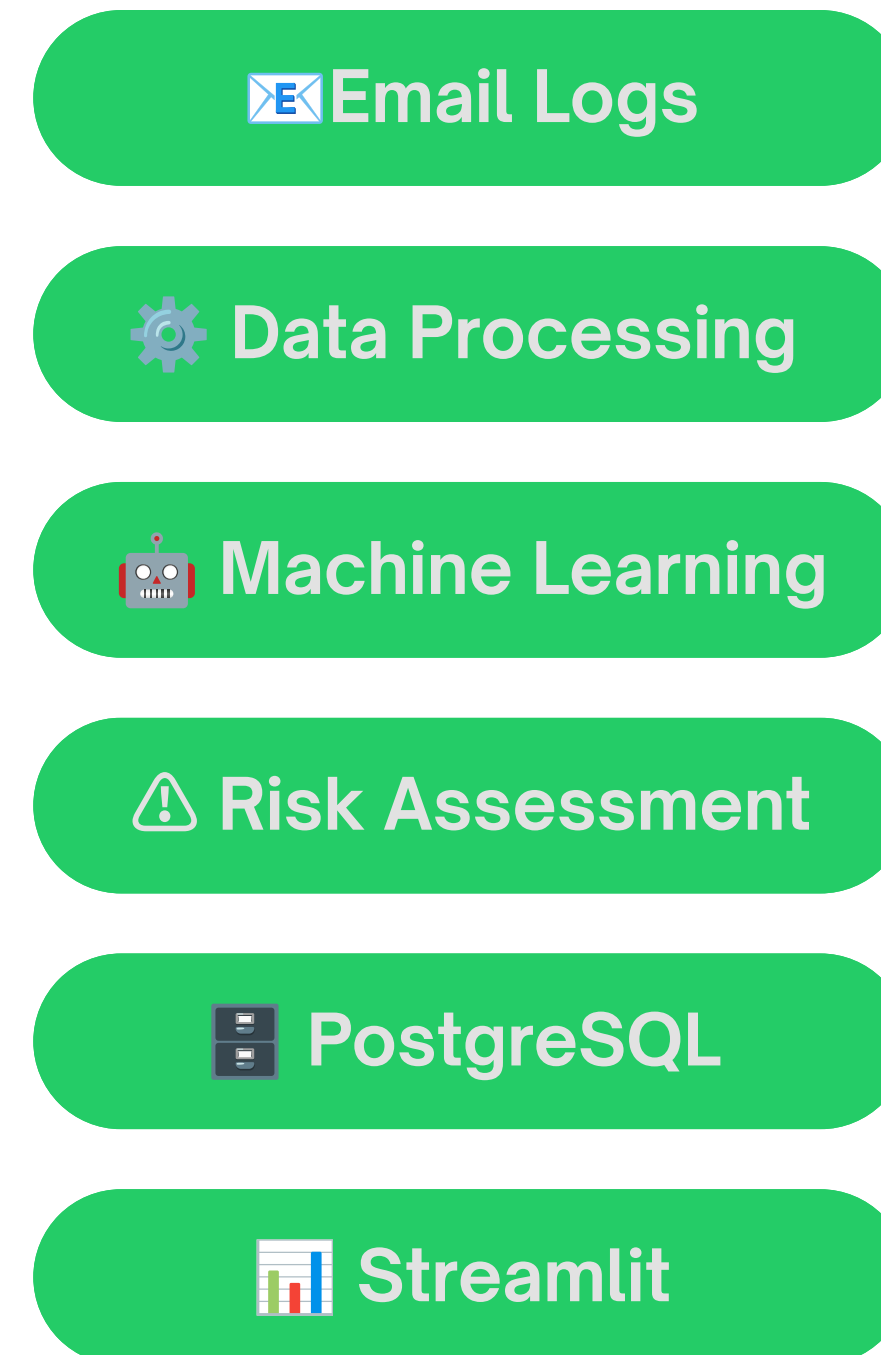
FROM RAW EMAIL DATA TO SECURITY INSIGHTS

Interactive reports
for managers

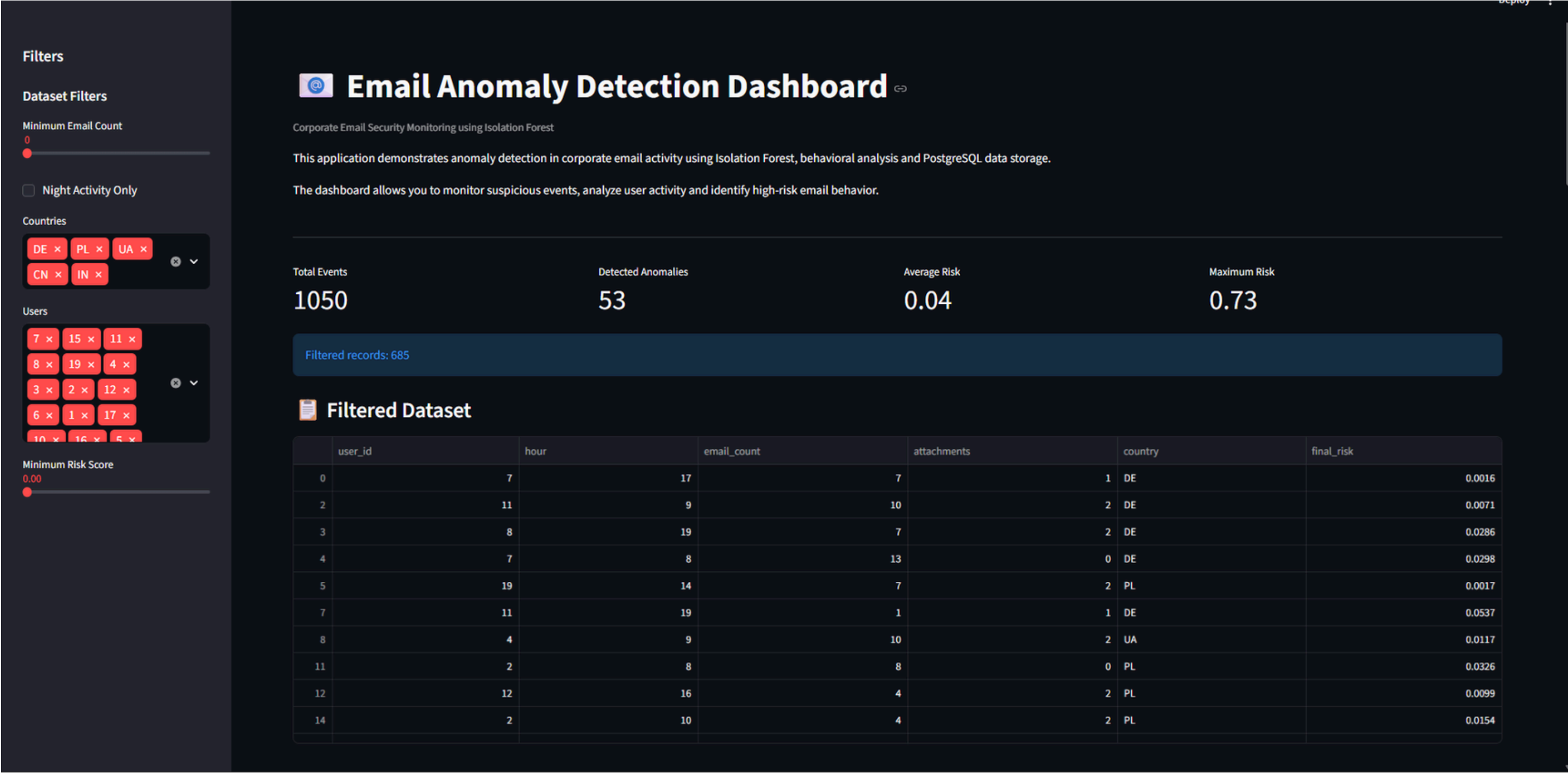
SYSTEM WORKFLOW



PROJECT PIPELINE



DASHBOARD OVERVIEW



Dashboard Capabilities:

- ✓ Event Monitoring
- ✓ Risk Scoring
- ✓ Country Analysis
- ✓ User Filtering
- ✓ Night Activity Detection
- ✓ CSV Export

MACHINE LEARNING PIPELINE

Data Generation

Synthetic corporate email activity

Machine Learning

Isolation Forest

Anomaly Detection

Feature Engineering

Country Encoding

Behavior Statistics

Risk Calculation

Behavior Score

Final Risk Score

RISK MONITORING

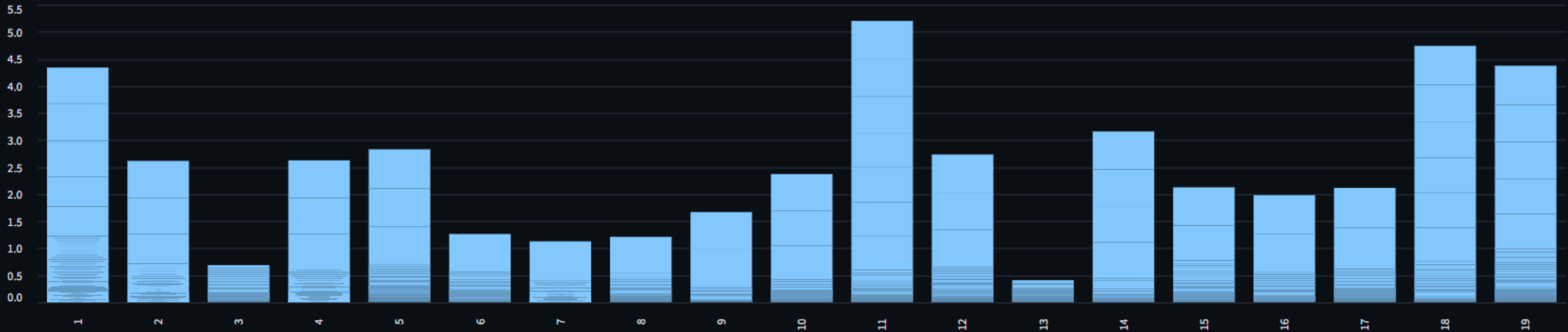
🔥 Top High-Risk Events 🔗

	user_id	hour	email_count	country	final_risk
1018	17	2	196	IN	0.7312
1034	7	1	175	CN	0.7309
1012	5	1	168	CN	0.7252
1032	19	2	180	CN	0.7225
1040	18	2	188	IN	0.7184
1042	17	3	185	IN	0.7162
1047	16	3	187	IN	0.7107
1028	5	4	187	CN	0.7068
1041	15	2	159	IN	0.7042
1007	16	3	166	CN	0.7032

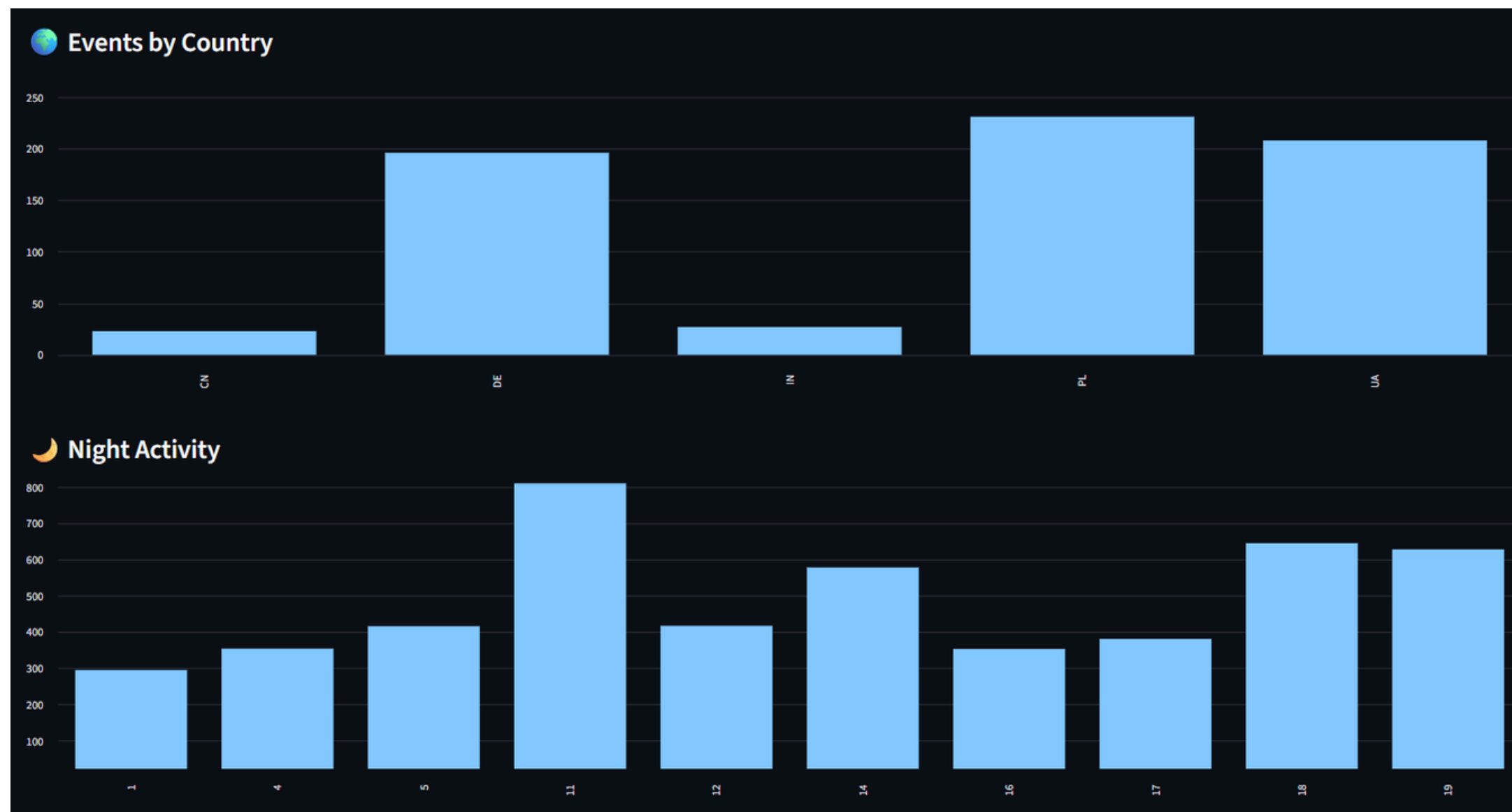
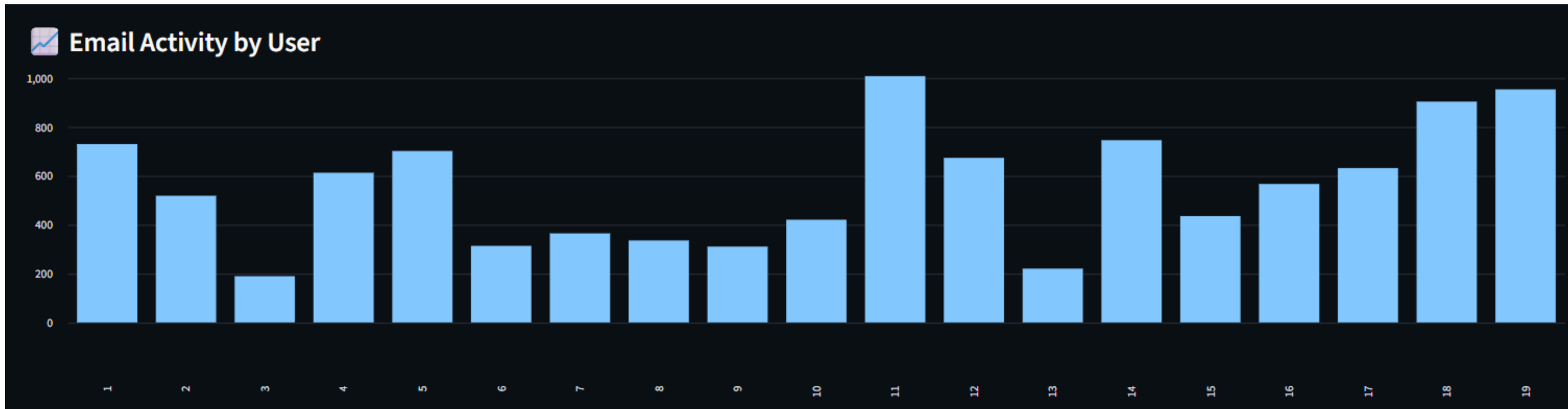
What can the analyst see?

- Highest-risk events
- Risk ranking
- Suspicious user behavior
- Prioritized investigation

📊 Risk Score Distribution



ACTIVITY ANALYTICS



Behavior Analysis

- Email volume by user
- Geographic distribution
- Night-time activity detection
- User behavior patterns

KEY INSIGHTS

NIGHT ACTIVITY

Night-time email activity is automatically highlighted as potentially suspicious.

BEHAVIOR ANALYSIS

Users with unusual email volume are identified automatically.

ANOMALY DETECTION

Isolation Forest detects suspicious behavior without labeled data.

RISK PRIORITIZATION

Events are ranked by final risk score.

SKILLS DEMONSTRATED

CORE TECHNOLOGIES

- Python
- Pandas
- Scikit-learn
- PostgreSQL
- Streamlit
- SQLAlchemy

KEY COMPETENCIES

- Machine Learning
- Behavior Analysis
- Risk Assessment
- Dashboard Development
- Data Visualization
- SQL

PROJECT RESULTS

1050

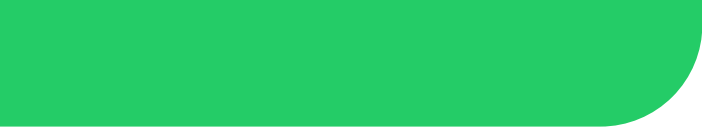
Generated Events

53

Detected Anomalies

Interactive Dashboard

CSV Export



THANK YOU!

[GITHUB](#)

[LINKEDIN](#)

CREATED BY
DMYTRO DIMURA

